Java 6 - Mergesort - Detyra të zgjidhura

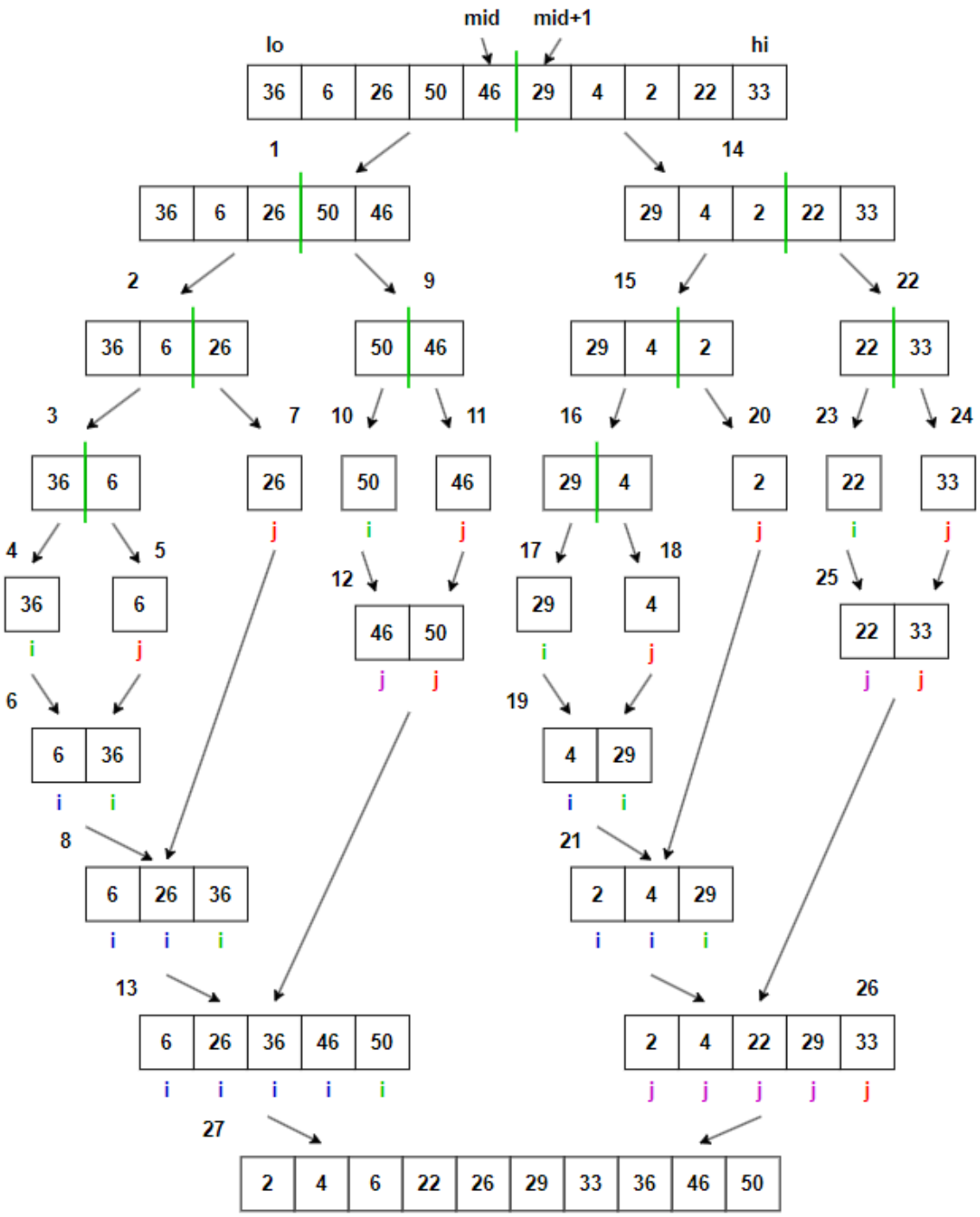
Lënda: Hyrje në Algoritme

Detyra 1. Të sortohet vargu i mëposhtëm duke përdorur algoritmin Merge Sort - Top Down

mid = lo + (hi - lo) / 2

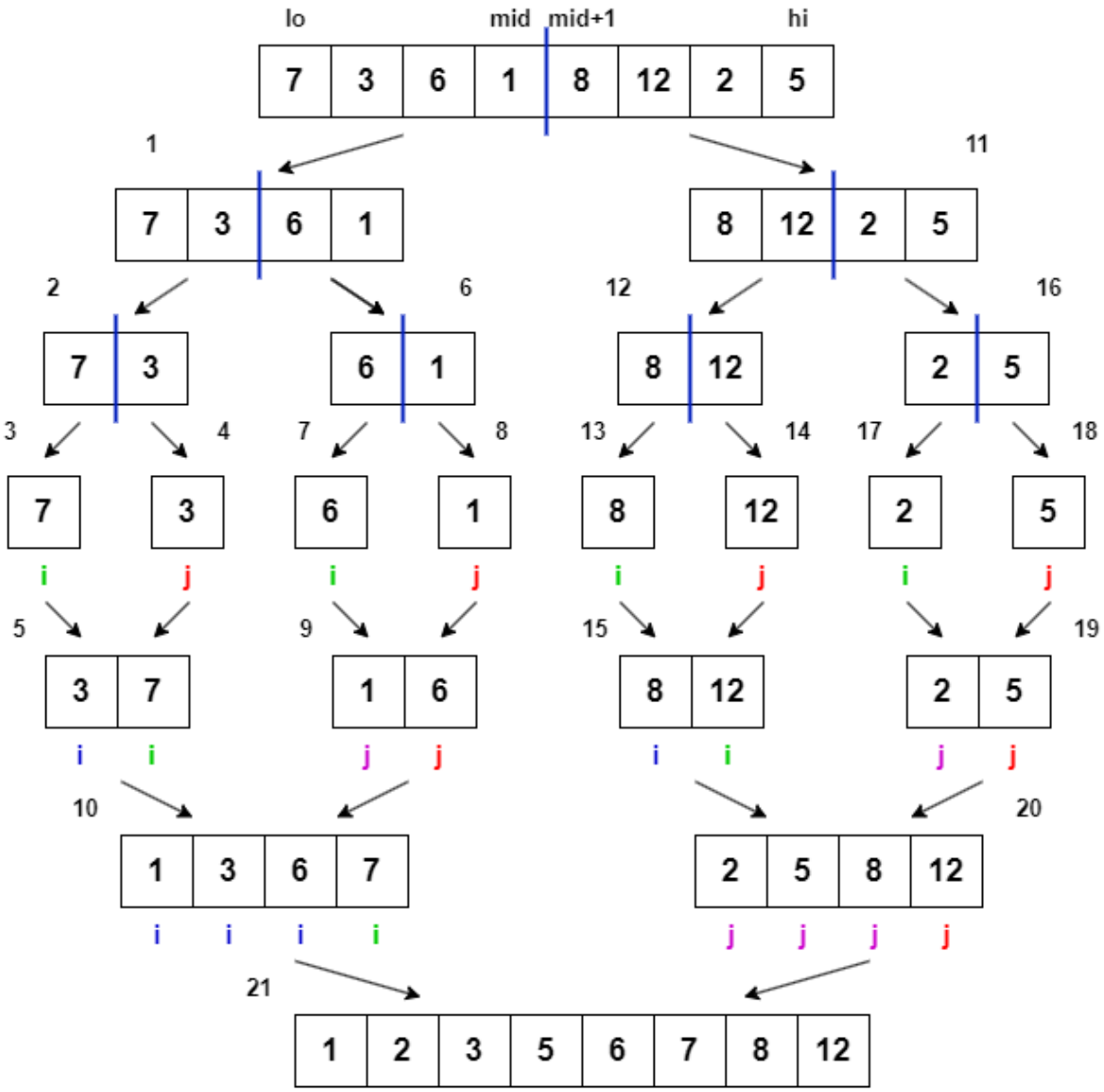
36	6	26	50	46	29	4	2	22	33	mid = 0+(9-0)/2=4.5
0	1	2	3	4	5	6	7	8	9	
lo				mid	mid+1				hi	
36	6	26	50	46						mid = 0+(4-0)/2=2
36	6	26								mid = 0+(2-0)/2=1
36	6									mid = 0+(1-0)/2=0.5
36										
i	6									
	j									
6	36									
i	i	26								
		j								
6	26	36								
i	i	i	50	46						mid=3+(4-3)/2=3.5
			50							
			i	46						
				j						
			46	50						
			j	j						
6	26	36	46	50						
i	i	i	i	i						
					29	4	2	22	33	mid=5+(9-5)/2=7

					29	4	2			mid=5+(7-5)/2=6
					29	4				mid=5+(6-5)/2=5.5
					29					
					i	4				
						j				
					4	29				
					i	i	2			
							j			
					2	4	29			
					i	i	i	22	33	mid=8+(9-8)/2=8.5
								22		
								i	33	
									j	
								22	33	
								j	j	
					2	4	22	29	33	
					j	j	j	j	j	
2	4	6	22	26	29	33	36	46	50	


mid
mid+1
lo
hi
36 6 26 50 46 29 4 2 22 33
1
14
36 6 26 50 46
29 4 2 22 33
2
9
15
22
36 6 26
50 46
29 4 2
22 33
3
7
10
11
16
20
23
24
36 6
26
50
46
29 4
2
22
33
4
5
12
17
18
25
36
6
46 50
29
4
22 33
6
19
6 36
4 29
8
21
6 26 36
2 4 29
13
26
6 26 36 46 50
2 4 22 29 33
27
2 4 6 22 26 29 33 36 46 50

Detyra 2. Të sortohet vargu i mëposhtëm duke përdorur algoritmin Merge Sort - Top Down

mid = lo + (hi - lo) / 2



Detyra 3. Të sortohet vargu a[] = 48, 27, 92, 17, 56, 97, 26, 86, 16, 23, 98 me anë të Merge Sort - Bottom-Up.

	48	27	92	17	56	97	26	86	16	23	98
Size 1	27	48	92	17	56	97	26	86	16	23	98
	i	j									
	27	48	17	92	56	97	26	86	16	23	98
			i	j							
	27	48	17	92	56	97	26	86	16	23	98
					i	j					
	27	48	17	92	56	97	26	86	16	23	98
							i	j			
	27	48	17	92	56	97	26	86	16	23	98
									i	j	
	17	27	48	92	56	97	26	86	16	23	98
Size 2	17	27	48	92	56	97	26	86	16	23	98
	i	i	j	j							
	17	27	48	92	26	56	86	97	16	23	98
					i	i	j	j			
	17	27	48	92	26	56	86	97	16	23	98
									i	i	j
	17	27	48	92	26	56	86	97	16	23	98
Size 4	17	26	27	48	56	86	92	97	16	23	98
	i	i	i	i	j	j	j	j			
	17	26	27	48	56	86	92	97	16	23	98
Size 8	16	17	23	26	27	48	56	86	92	97	98
	i	i	i	i	i	i	i	i	j	j	j
	16	**17**	**23**	**26**	**27**	**48**	**56**	**86**	**92**	**97**	**98**

Detyra 4. Të sortohet vargu a[] = G, Q, D, R, Z, S, X, E, Y, P me anë të merge sort - Bottom-Up.

G	Q	D	R	Z	S	X	E	Y	P

Size = 1

G	Q	D	R	Z	S	X	E	Y	P
i	j								
G	Q	D	R	Z	S	X	E	Y	P
		i	j						
G	Q	D	R	S	Z	X	E	Y	P
				i	j				
G	Q	D	R	S	Z	E	X	Y	P
						i	j		
G	Q	D	R	S	Z	E	X	P	Y
								i	j

Size = 2

G	Q	D	R	S	Z	E	X	Y	P
D	G	Q	R	S	Z	E	X	Y	P
i	i	j	j						
D	G	Q	R	E	S	X	Z	Y	P
				i	i	j	j		

Size = 4

D	G	Q	R	E	S	X	Z	Y	P
D	E	G	Q	R	S	X	Z	Y	P
i	i	i	i	j	j	j	j		

Size = 8

D	E	G	Q	R	S	X	Z	Y	P
D	E	G	P	Q	R	S	X	Y	Z
i	i	i	i	i	i	i	i	j	j

Detyra 5. Të sortohet vargu a[] = 1 ,3 ,4 ,7 ,10 ,2 ,4 ,7 ,1 ,4 ,6 ,10 me anë të merge sort - TimSort

1 3 4 7 10 2 4 7 1 4 6 10

1 3 4 7 10 2 4 7 1 4 6 10

1 3 4 7 10 2 4 7 1 4 6 10

i i i i i j j j

Merge

1 2 3 4 4 7 7 10 1 4 6 10

1 2 3 4 4 7 7 10 1 4 6 10

i i i i i i i i j j j j

Merge

1 1 2 3 4 4 4 6 7 7 10 10

Detyra 6. Të sortohet vargu a[] = 4 ,5 ,3 ,7 ,10 ,1 ,5 ,8 ,9 ,2 ,12 ,10 me anë të merge sort - TimSort

4 5 3 7 10 1 5 8 9 2 12 10

4 5 3 7 10 1 5 8 9 2 12 10

4 5 3 7 10 1 5 8 9 2 12 10

i i j j j

Merge

3 4 5 7 10 1 5 8 9 2 12 10

3 4 5 7 10 1 5 8 9 2 12 10

3 4 5 7 10 1 5 8 9 2 12 10

Merge

3 4 5 7 10 1 2 5 8 9 10 12

i i i i j j j

3 4 5 7 10 1 2 5 8 9 10 12

3 4 5 7 10 1 2 5 8 9 10 12

i i i i i j j j j j j j

Merge

1 2 3 4 5 5 7 8 9 10 10 12